Combined Safety and Security Analysis in Self-Adaptive Systems

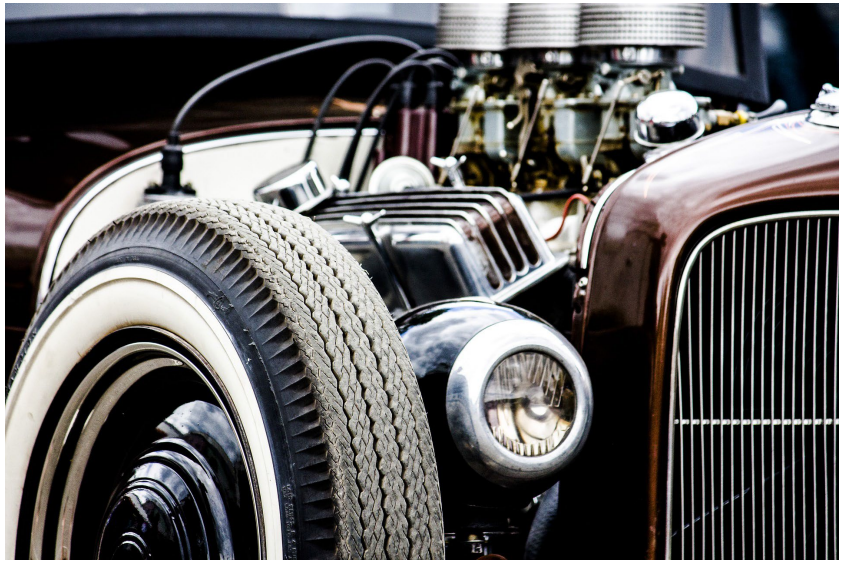
SafeSec Workshop – Introduction and Agenda

Workshop Agenda

16:00 – 16:05	*– Introduction –*	
16:05 – 16:20	*Talk:*	SAFT-GT Overview and Agenda
16:20 – 16:30	*Discussion:*	SAS: Experiences & Challenges
16:30 – 16:50	*Talk:*	Input Models and Model Generation
16:50 – 17:10	*Discussion:*	Safety and Security Modeling and Analysis
17:10 – 17:15	*Pause*	
17:15 – 17:35	*Talk:*	AFT Model Combination and Analysis
17:35 – 17:55	*Discussion:*	Impressions and Comments on the Prototype
17:55 – 18:00	*– Closing –*	

Self-Adaptive Systems (SAS)

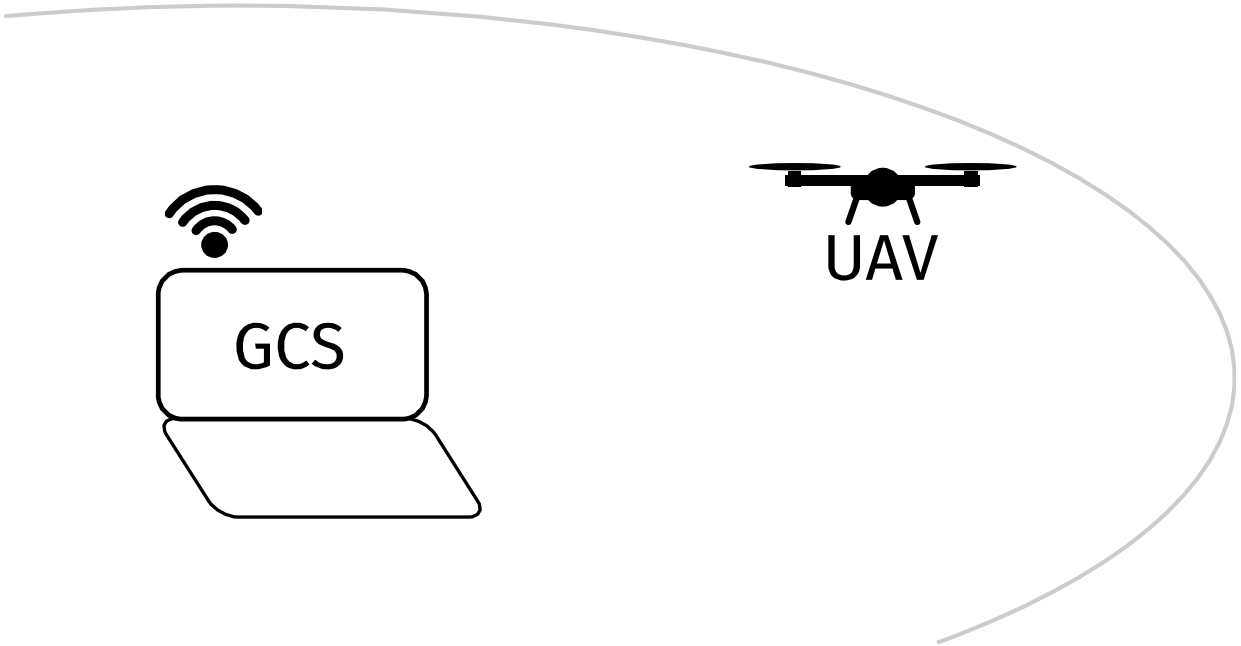
- Discontinuous behavior changes at run-time in reaction to environment changes
- Ranges from parameter changes to architectural reconfiguration
- Separates adaptation rules from behavior
- Safety and Security properties may change



ROS Demonstrator System

- Quadcopter system with optical tracking
- Components can be activated/deactivated
- System monitoring with ROS graph introspection capabilities
- Managing MAPE-K System implemented in ROS

Combined Safety and Security Analysis Example



WiFi connected > Radio Connected > Return signal > WiFi connected

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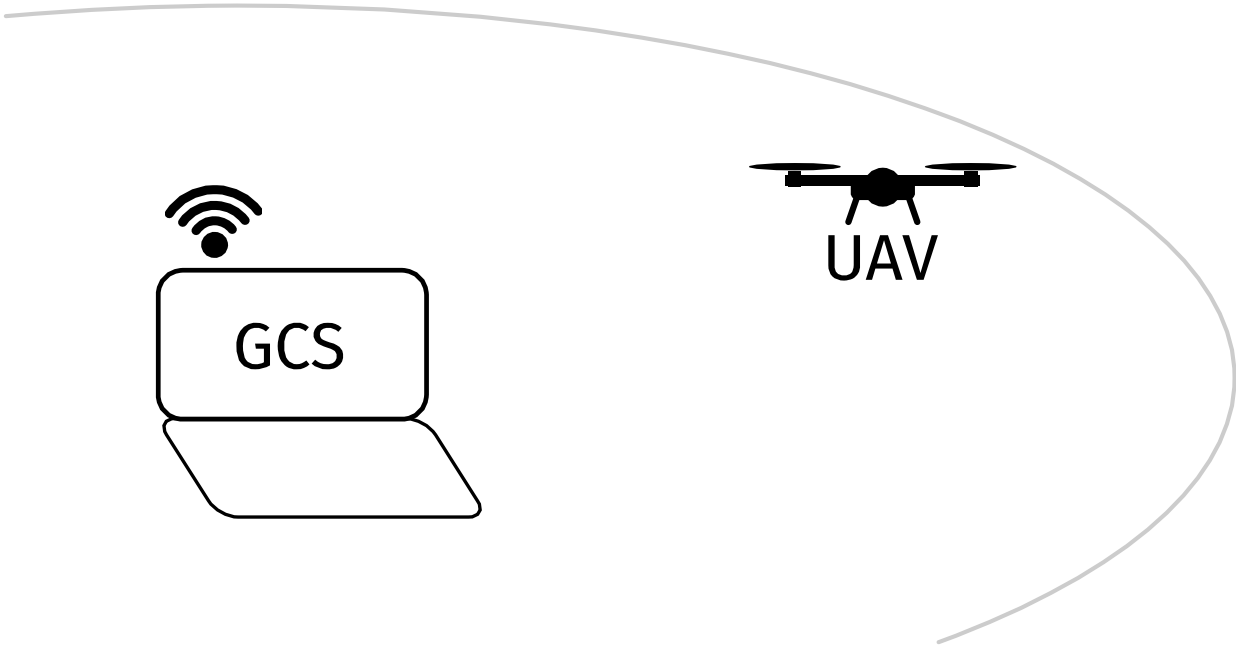

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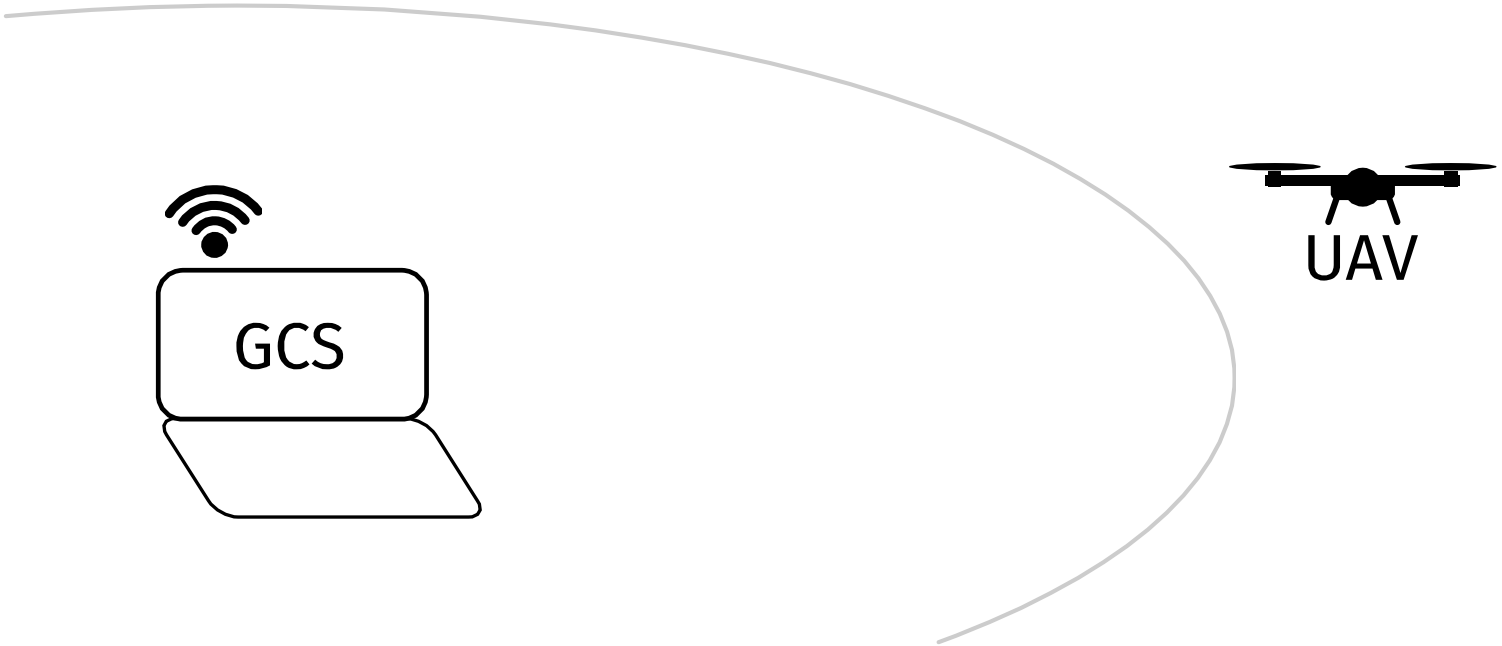

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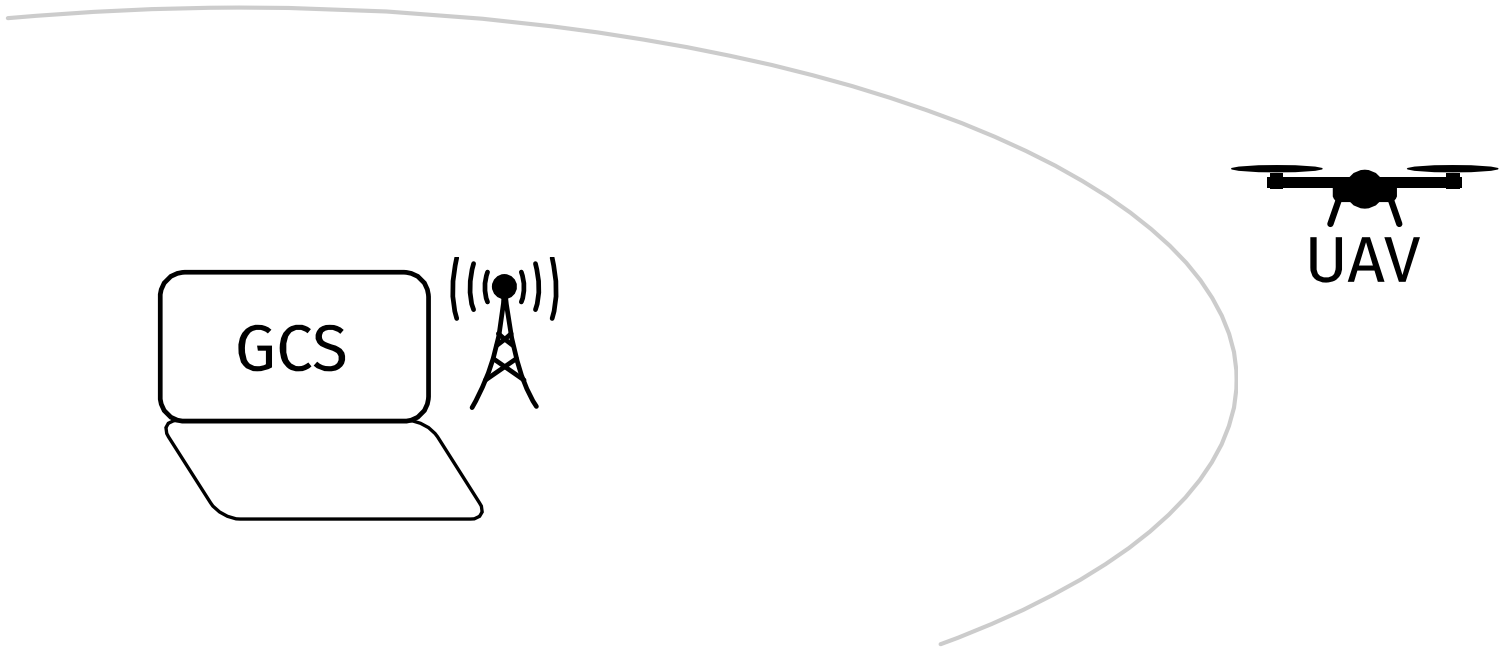

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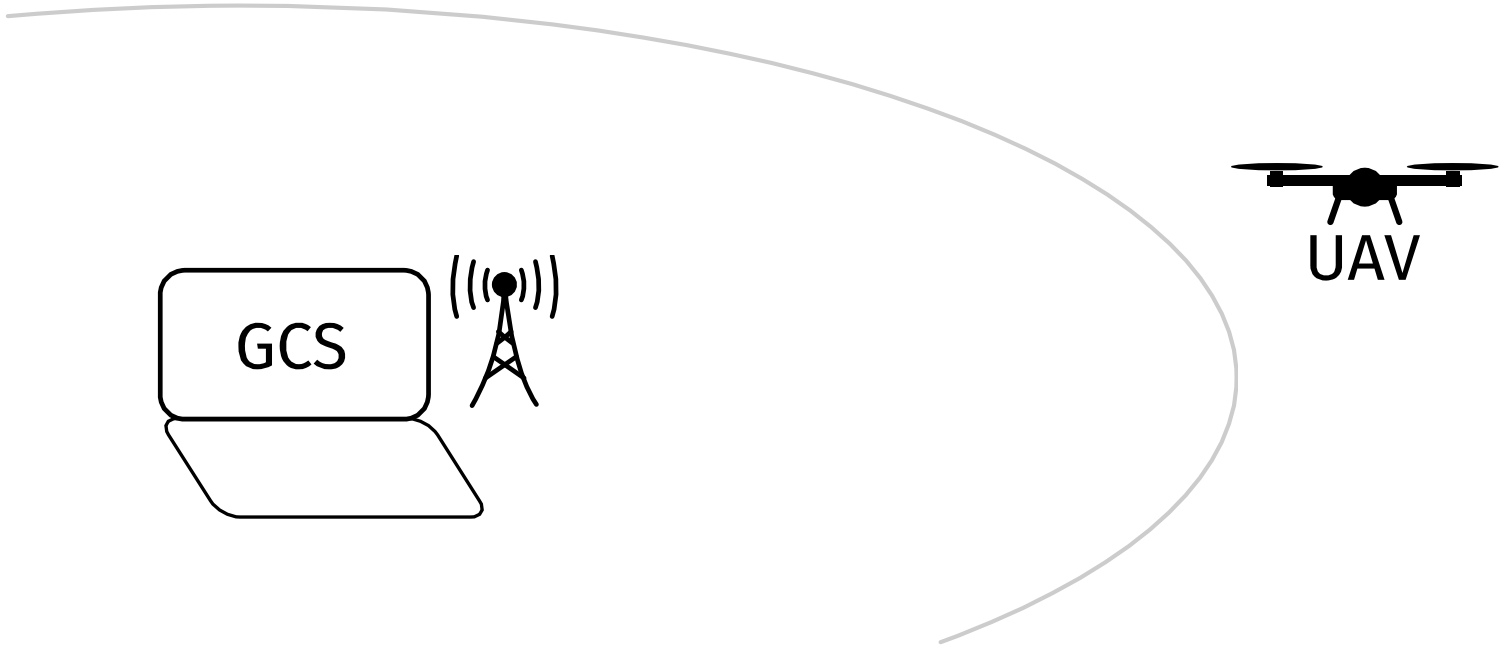

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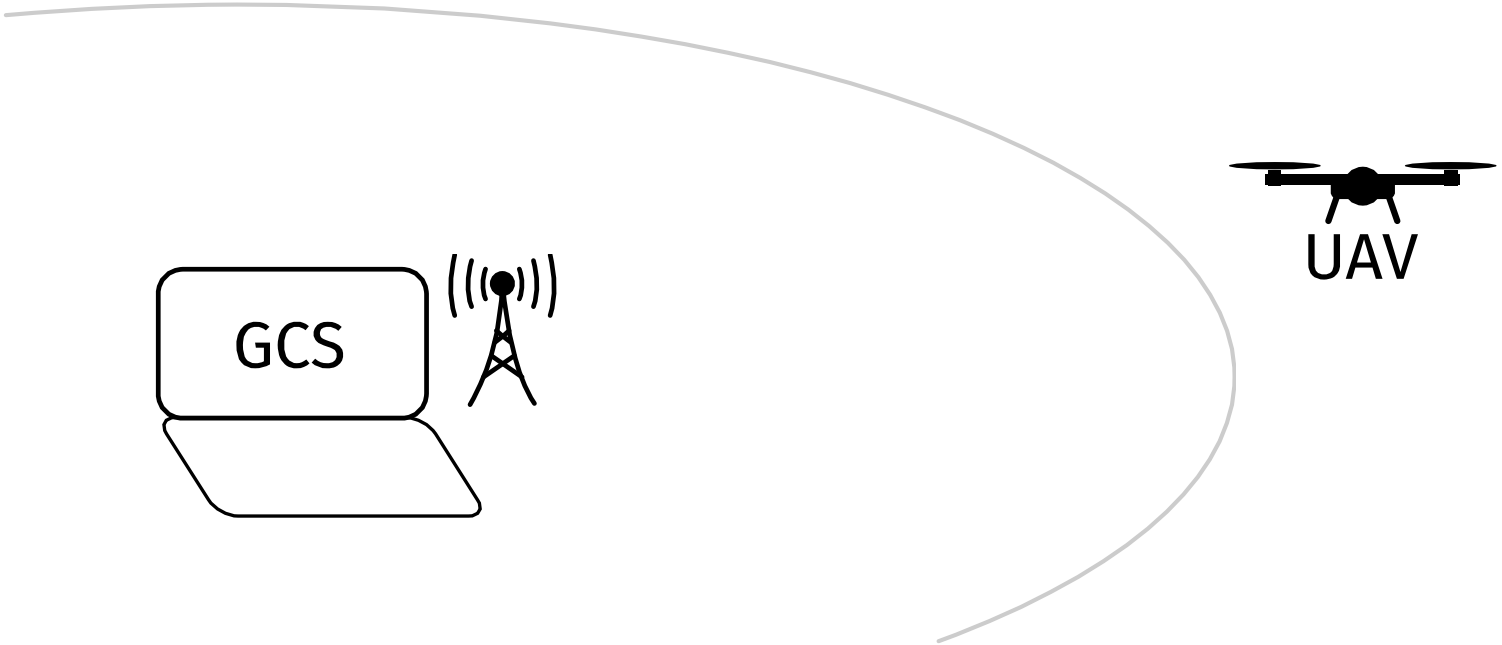

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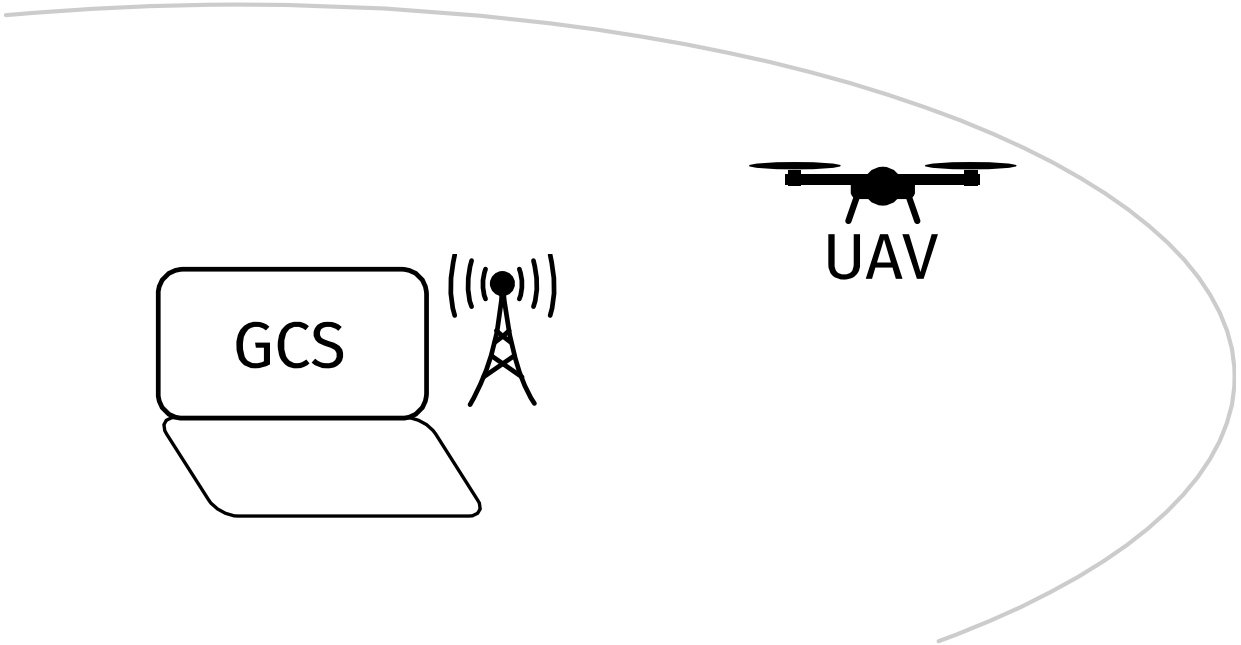

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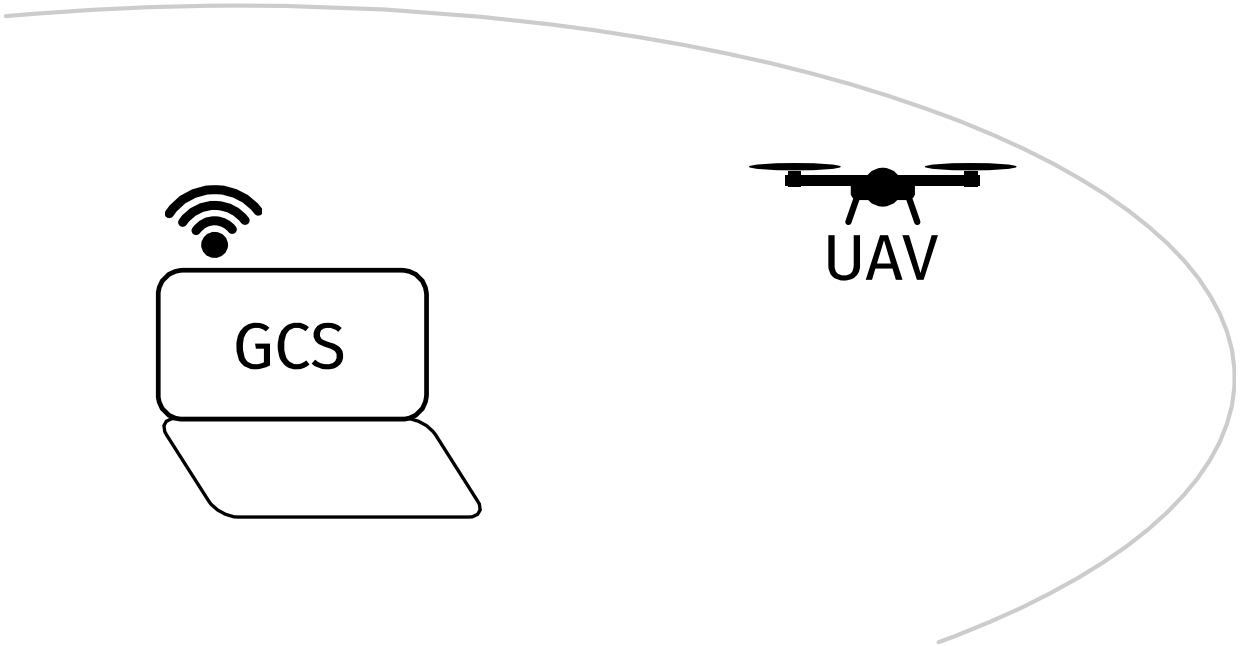

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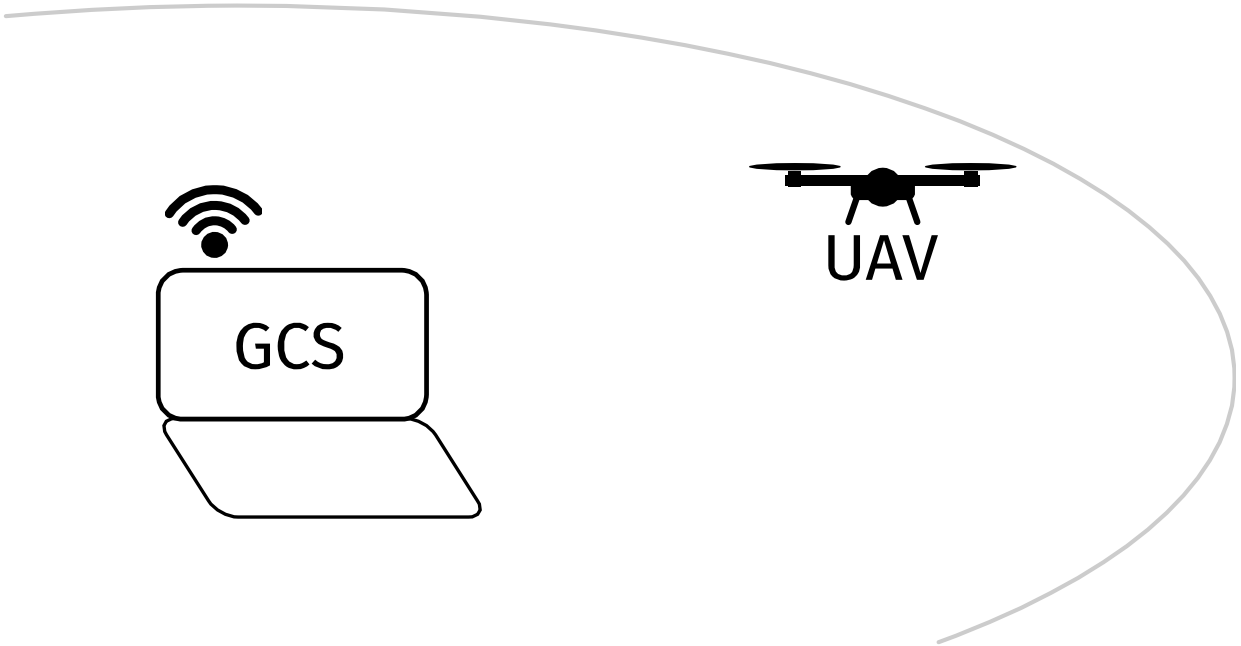

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Our Vision:
The SafeSec AFT Generation Toolchain (SAFT-GT)

Automated input

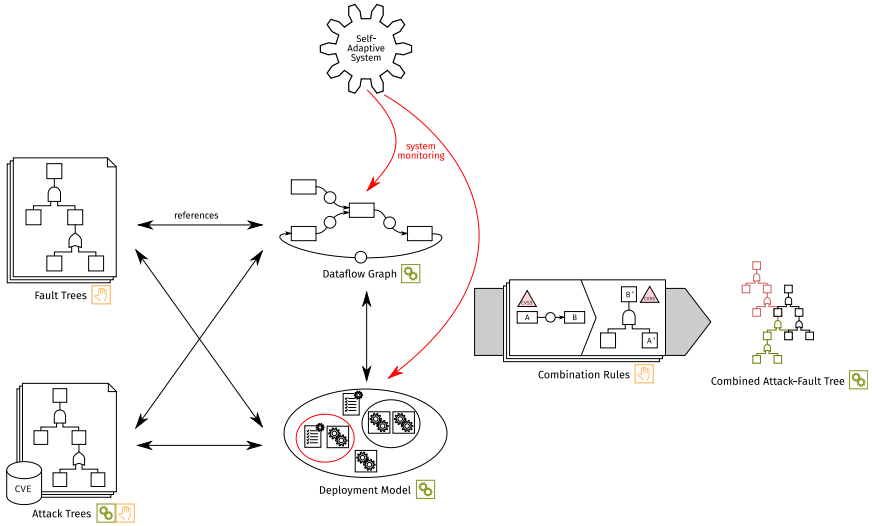
- ATs from mined vulnerabilities
- Dataflow model
- Deployment model

Manual input

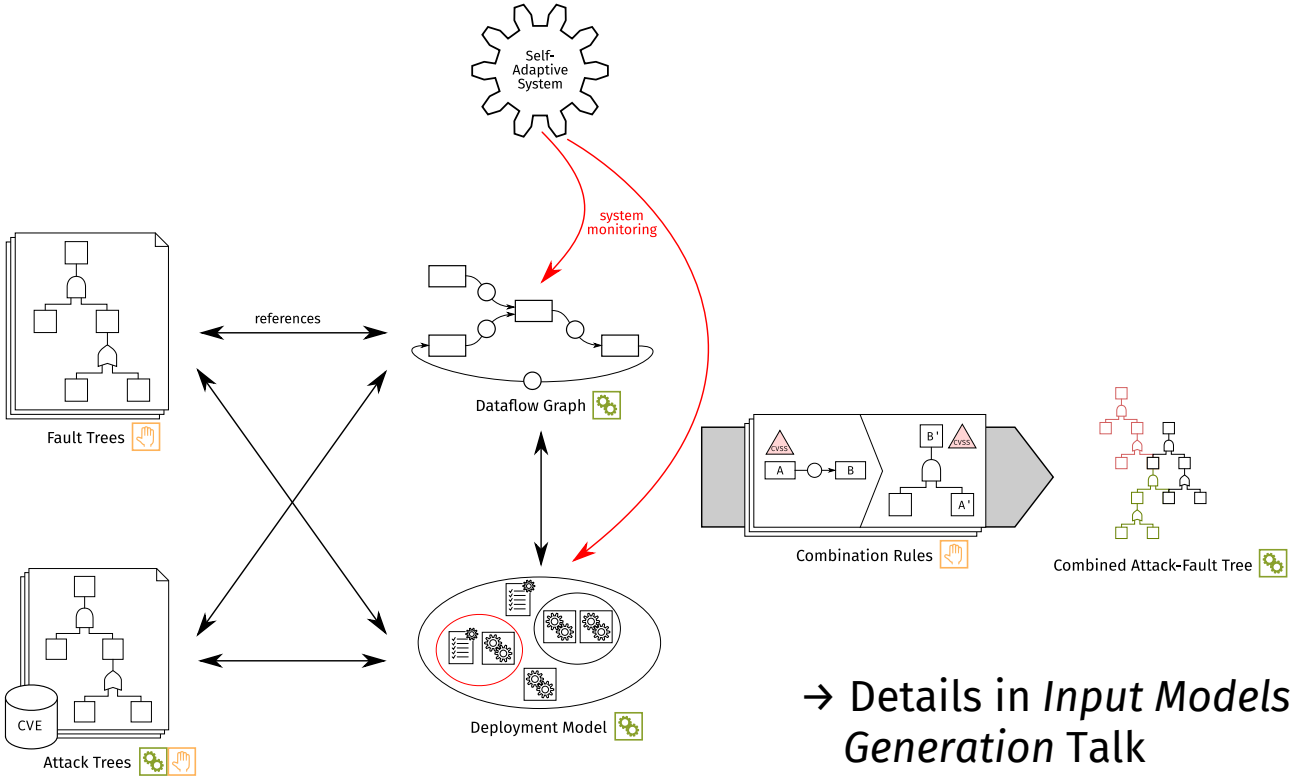
- Static fault trees
- common generic attack patterns/translation rules

Output

- generated AFT



Workshop Agenda – Modeling

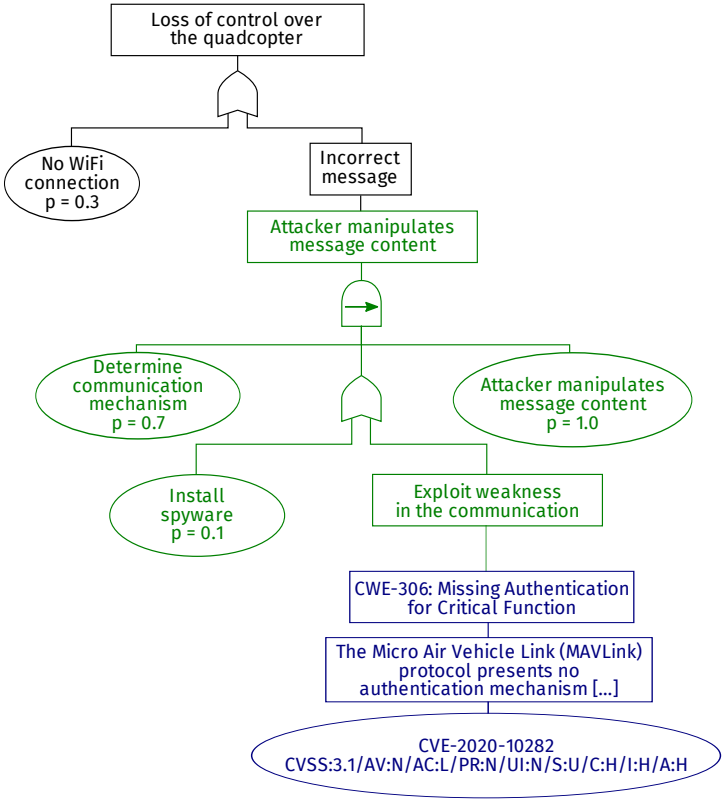


→ Details in *Input Models and Model Generation* Talk

Workshop Agenda – Model Combination

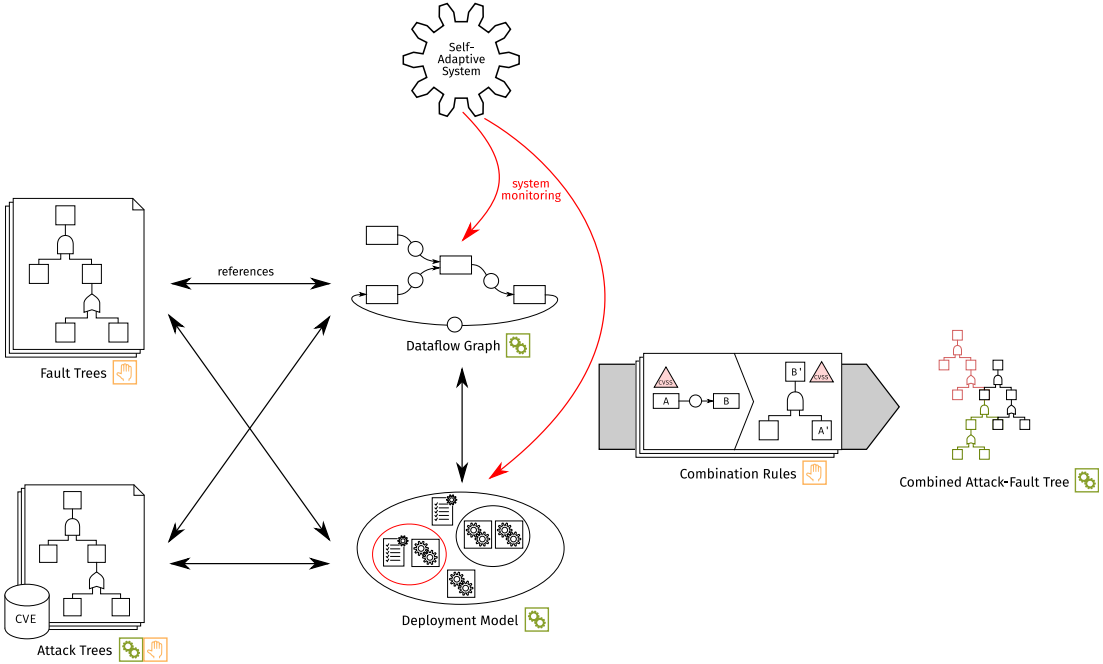
- Combine ATs and FTs into AFTs automatically
- Rule-based approach, based on patterns in the input models
- Allows continuous model regeneration after changing the application composition

→ Details in *AFT Model Combination and Analysis* Talk



Thank You.

Any Questions?



→ Next up: SAS and Safety/Security Analysis Discussion